Concordia University

SOEN 471 - Big Data Analytics
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Assignment 1
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Theoretical Analysis

Decision trees are well suited for predicting customer churn because they can effectively model complex, non-linear relationships between customer behaviour and churn outcomes. In this case, churn is likely influenced by multiple factors such as complaints, payment issues, resolution delays, etc. Decision trees naturally capture these interaction effects through hierarchical splits without requiring assumptions about linearity.

Moreover, the dataset contains both numerical and categorical variables. Decision trees can handle such mixed data types efficiently after basic preprocessing, thus making them practical for such datasets. Another key advantage is interpretability. Decision trees generate 'if-then' rules (e.g., if complaints are high and resolution time is long, then churn risk increases) which are easy for managers and stakeholders to understand. In a business context like StreamFlex, interpretability is essential because predictive insights must translate into actionable strategies.

Decision trees combine strong analytical power with practical interpretability. They capture complex customer churn patterns and translate them into clear, actionable rules for informed business decision-making. One of the primary advantages of Decision Trees in a business context is their high level of interpretability and transparency. Unlike complex "black-box" models such as neural networks, Decision Trees function as "white-box" models, meaning their decision-making process can be easily visualized and understood by non-technical stakeholders. For a company like Stream Flex, this is crucial because management and marketing teams need to trust the logic behind a churn prediction before allocating budget to retention efforts. The model provides a clear, logical flowchart that explains exactly why a specific customer was classified as "at risk," which fosters confidence in the results and facilitates easier adoption across the organization.

Furthermore, Decision Trees are highly effective at generating actionable business rules. The model naturally segments data into specific groups using "if-then" logic (e.g., "If subscription is Basic and payment has issues, then churn is likely"), which allows the business to identify the exact conditions leading to customer loss. This feature is particularly useful for the provided dataset, which contains mixed data types like Membership_Type and Payment_Method. By automatically selecting the most important features to split on, the Decision Tree highlights the key drivers of churn, allowing Stream Flex to prioritize its resources on the most significant friction points such as fixing payment systems or engaging specific demographic groups rather than relying on intuition alone.

Churn predictions can be used by StreamFlex to improve customer retention because they allow for customers who are at a higher risk of canceling their subscription to be identified before

they actually leave. This allows the company to recommend content tailored to their preferences or be proactive in reaching out to customers who may be unhappy with the service to offer personalised discounts. For example, customers who have made complaints multiple times are classified as high risk, StreamFlex can prioritize improving the customer support quality and addressing those complaints specifically. By addressing these issues early there will likely lead to a reduction in revenue loss and an increase in customer retention.

In addition, churn predictions can also be used by StreamFlex to design targeted marketing strategies by focusing on high-risk customers instead of applying equal efforts to all. Customers with low watch times can receive content recommendations based on what they have previously enjoyed. Any customer that experiences payment issues can be offered more flexible payment options as well, for example, biweekly payments instead of monthly payments to charge smaller amounts at a time. These targeted strategies improve marketing efficiency because the needed attention and resources are put towards customers who are more likely to churn.

Lastly, to enhance customer support operations, churn predictions can be used by StreamFlex to reduce churn if Number_of_Complaints and Resolution_Time_Days show to have influence. Since these are directly correlated with service quality and customer experience on the platform StreamFlex can help to reduce response time and improve the resolution process for better customer experience. It can also be used to prioritize customers who have a higher level of subscriptions as well, so that high paying customers are at less churning risk. Churn predictions help customer support operations because it uses data based improvements towards customer service enhancement.

Methodology

The dataset was first examined for data quality issues. No missing values and duplicate rows were detected, and no invalid entries were found in the features. The 'CustomerID' column was checked to ensure each record represented a unique customer. as seen in Image 1 and Image 2.

To improve interpretability and better structure the model, shown in Image 3, input for training, selected numerical variables were converted into ordered numerical classes: 'Age' was grouped into six ordered categories based on its range (18-69) and 'Watch_time_hours" was divided into five usage levels representing increments of 50 hours.

Categorical values were encoded into numerical form to ensure compatibility with scikit-learn models: 'Payment_Method' was mapped into integer labels, 'Preferred_Content_Type' was mapped into numerical categories, 'Membership_Type' was converted into ordered levels.

The dataset was then separated into input features (X) and target features (y), as shown in Image 4, with dropped 'CustomerID' since it does not contribute predictive value. The data was split into training (70%) and testing (30%) sets using 'train_test_split' with 'random_state=42' to ensure reproducibility.

For decision tree model training and hyperparameter tuning, a 'DecisionTreeClassifier' was used as the base model for churn predictions. The model was initialized with 'random_state=42'. To optimize performance, 'GridSearchCV' with 5 fold cross validation was applied to training data. The following hyperparameters were explored: criterion (measures the split quality), max_depth, (controls tree depth to reduce overfitting), Min_samples_split (minimum samples required to split a node), and Max_features (number of features considered at each split). The grid_search used 'scoring="accuracy"' to identify the parameter combination yielding the highest cross-validated accuracy and the final model used for evaluation was 'grid_search1.best_estimator_'. This implementation is shown in Image 5.

As displayed in Image 6, the optimized Decision Tree model was evaluated on the unseen test set. Predictions generated using the predict function. Next, the following metrics were computed: accuracy (overall proportion of correct predictions), precision (proportion of predicted churners who actually churned), recall (proportion of actual churners correctly identified), F1-score (harmonic mean of precision and recall), and confusion matrix (counts of True Positives, True Negatives, False Positives and False Negatives). These metrics are chosen to observe a complete view of performance, since the churn dataset is imbalanced.

For random forest model training and hyperparameter tuning, to improve task performance, a 'RandomForestClassifier' was trained using the same training dataset. A grid search on the following hyperparameters was observed: N_estimators (number of trees), Max_depth (tree depth), Min_samples_split (split threshold), and Max_features (number of features per split). GridSearchCV with 5 fold cross-validation and 'n_jobs=-1', for parallel processing, were used to efficiently search the parameter space. The best-performing configuration 'grid_search2.best_estimator_' was selected based on cross validation accuracy. This implementation is shown in Image 7.

As displayed in Image 8, the optimized Random Forest model was evaluated on the unseen test set. Predictions generated using the predict function. Next, the same metrics as previously were computed, including accuracy, precision, recall, F1-score, and confusion matrix. Similar to the decision tree model, these metrics are chosen to observe a complete view of performance, since the churn dataset is imbalanced.The objective of feature evaluation is to quantify the contribution of each predictor feature to churn classification. Multiple evaluation techniques were used to ensure edge cases and to compare feature influence across different

metrics: Gini Index of each feature, Information Gain of each feature, and Feature Importance for the two models.

Gini Index of each feature, measures node impurity in classification trees. The minimum value, 0, means when a node produces a pure split (i.e., all samples belong to one class). The maximum value, **0.5 (in this code block, the value is 1)**, means when classes are perfectly balanced and impure. Features contributing to greeting impurity decrease are considered more influential classification. Information Gain of each feature, measures the reduction in entropy after splitting based on a feature.

Information Gain of each feature identifies features that provide the greatest reduction in classification uncertainty. In churn prediction, features with high information gain strongly differentiate churned from retained customers.

Feature Importance based on both classifiers trained on training set: feature importance was extracted from the best performing model using model_name.best_estimator_.feature_importances_ . In tree-based models, feature importance is computed as the normalized total reduction of impurity contributed by each feature across all splits in the tree (and across all trees in the Random Forest). Feature Importance shows how much feature influences the structure of that single tree. In Random Forest, Feature Importance is averaged across many trees, providing a more stable and variance-reduced estimate of feature contribution.

To conclude, Gini and Information Gain were used for theoretical assessment, and Feature Importance for empirical assessment of real model behaviour. This approach reduces relying on a single criterion.

Results

The Decision Tree classifier was optimized using GridSearchCV with 5-fold cross-validation on the training set. The best hyperparameters were when evaluated on the test set, the tuned Decision Tree model obtained that the confusion matrix indicates that the model correctly identifies the majority of churned customers (high recall), while misclassifying 31 non churned cases, and 14 churned cases.

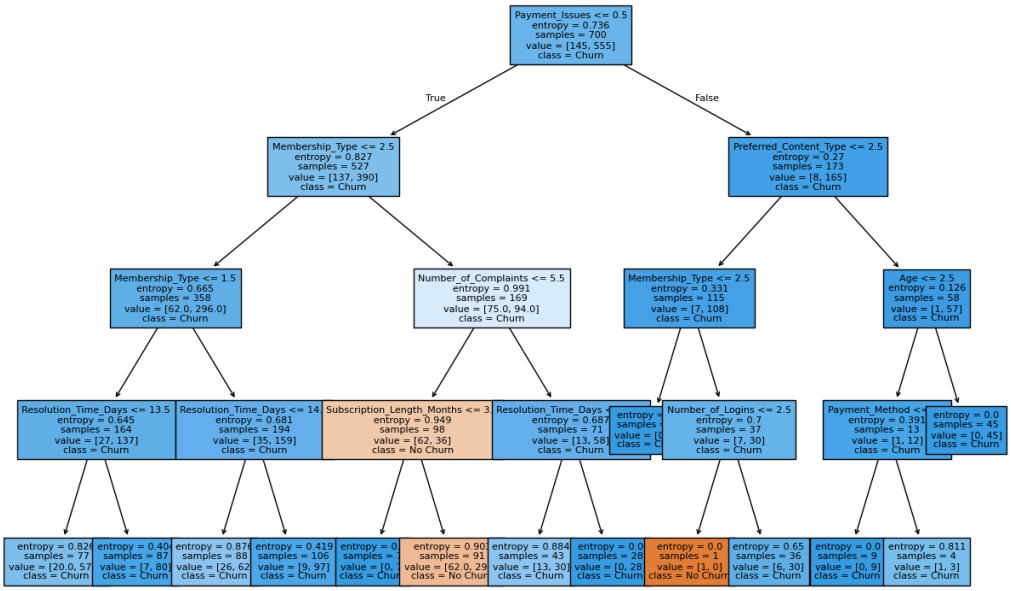

Figure 1: Decision Tree Classifier Model Plot.

The Random Forest Classifier was similarly tuned using GridSearchCV with 5-fold cross-validation over a broader hyperparameter grid. The best hyperparameters were when evaluated on the test set, the tuned Random Forest model obtained a confusion matrix which indicates that the model correctly identifies a greater proportion of churned customers (highest recall among the two models), although it produces slightly more false positives than the Decision Tree.

Table 1: Comparison of the two classifier models.

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.850	0.878	0.941	0.909
Random Forest	0.843	0.860	0.958	0.907

Overall, both models perform similarly with nearly identical F1-Scores. The Decision Tree achieves marginally higher accuracy and precision, indicating fewer false churn predictions. In contrast, the Random Forest achieves higher recall, meaning it is more effective at identifying customers who actually churn. In churn prediction, recall is often a critical metric because failing to identify a churner may result in lost revenue. From this perspective, the Random Forest offers a slight practical advantage, despite similar overall performance.

When it comes to feature importance while using Gini Index, the most influential variables are Membership_Type, Payment_Issues, Resolution_Time_Days and Number_of_Complaints. These variables dominate the tree's splitting decisions. Other features contribute less directly to the tree's predictions. When using Information Gain, the most influential features are Number_of_Complaints, Payment_Issues, Resolution_Time_Days, Payment_Method, and Membership_Type. However, Subscription_Length_Months and Number_of_Logins showed near zero Information Gain, which suggests that even though these variables have high Gini Index, they may not significantly reduce entropy. The strongest entropy reducing feature is Number_of_Complaints.

1. Feature Importance (Decision Tree Classifier):
   The Decision Tree relies mostly on Membership_Type, Payment_Issues, Resolution_Time_Days and Number_of_Complaints.
   Watch_Time_Hours contributes almost nothing in the final tree, indicating it was not selected in major splits. Figure 19 visually confirms this dominance.

2. Feature Importance (Random Forest Classifier):
   Unlike the Decision Tree Classifier, the Random Forest Classifier does not rely excessively on one dominant variable. It captures broader interactions across behavioural, temporal and demographic features.

   The most influential features are Number_of_Complaints, Subscription_Length_Months, Resolution_Time_Days, and Number_of_Logins.

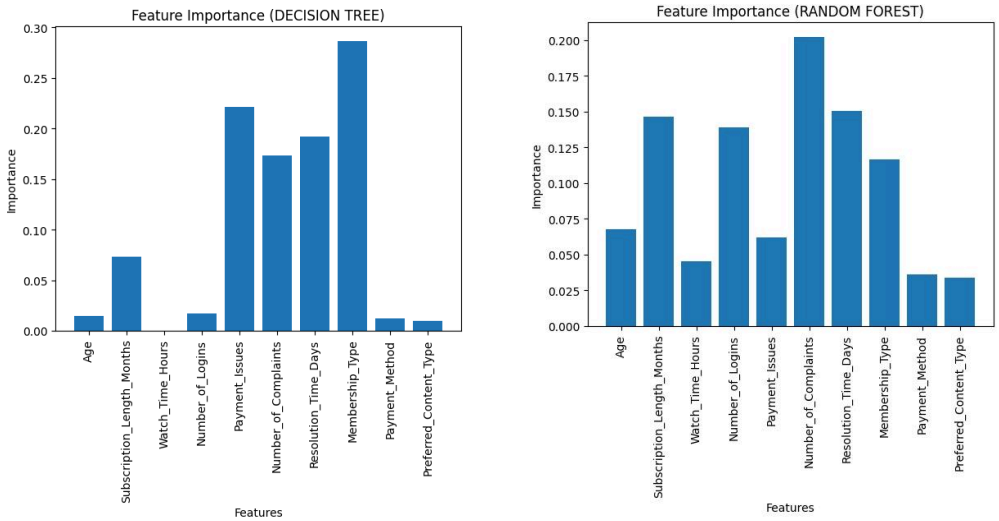

Figure 2: Feature Importance Bar Plot Based on Decision Tree Classifier and Random Forest Classifier.

In comparison, across both models, the features Number_of_Complaints, Membership_Type, Subscription_Length_Months, Resolution_Time_Days, and

Number_of_Logins are influential. The Decision Tree model concentrates importance in a small subset of dominant features. The Random Forest model distributes predictive influence across more variables, presenting the key difference between a high variance single-tree model and a variance-reduced forest model.

Visualizations of Exploratory Data Analysis (EDA):
To better understand the structure of the dataset and identify patterns, several visualizations were created. These plots provide insights into the demographic distribution, behavioural trends, class imbalance, and relationships between features and churn.

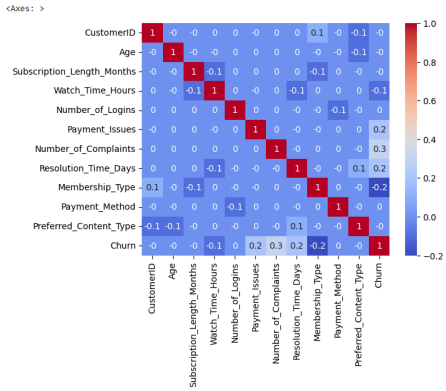

Figure 3: Correlation Heatmap.

The dataset consists of 1,000 observations and 12 features, including numerical features, categorical features, and a binary target column (Churn). Data quality assessment confirms that there are no missing values and no duplicate records, ensuring reliability for downstream analysis. The correlation heatmap, Figure 3, illustrates pairwise relationships among numerical features. Higher correlation observed between Number_of_Complaints and Churn, Resolution_Time_Days and Churn, and lastly Payment_Issues and Churn. No significant multicollinearity is observed among predictors, indicating that the features provide relatively independent information to the models.

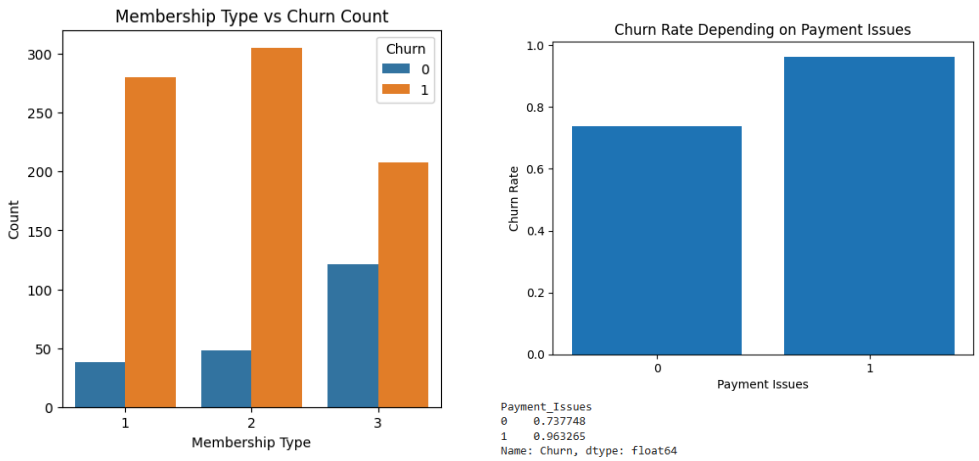

Figure 4: Churn Frequency Based on Membership Type and Payment Issues.

The Standard tier shows relatively higher churn proportions. This aligns with the feature importance results from the Decision Tree model, where Membership_Type was the most influential predictor. The pricing and perceived value of the tiers might influence retention. The dataset is imbalanced, with churn representing the majority class. Therefore, evaluation metrics such as Recall, Precision and F1-Score provide more insightful values than Accuracy alone, as accuracy can be inflated in imbalanced settings.

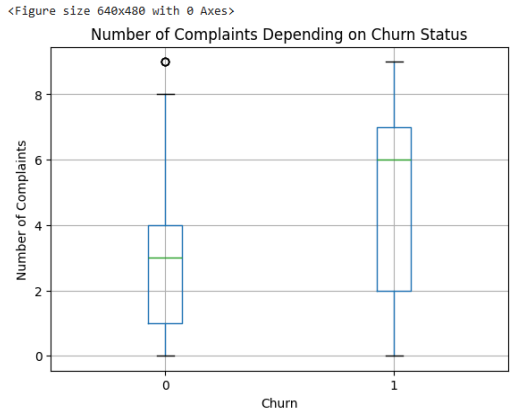

Figure 5: Probability of Churning Based on Number of Complaints.

To conclude, the EDA reveals that churn behaviour is primarily driven by customer dissatisfaction and service related issues, rather than demographic factors alone. To highlight, high complaint frequency, longer issue resolution times, payment-related problems and perceived value of membership types are consistently associated with increased churn risk. Engagement metrics and content preferences also contribute, though to a lesser extent, suggesting that behavioural patterns interact with service experience in influencing retention.

Business Insights

According to the data, operational and service-related factors, rather than demographic traits are the main causes of customer attrition at StreamFlex. The most significant predictors in both models and feature evaluation techniques are Number_of_Complaints, Resolution_Time_Days, Payment_Issues, and Membership_Type. These results align with the report's feature importance analysis and EDA results. Customer dissatisfaction emerges as the strongest churn driver. Consumers are much more likely to terminate their subscriptions if they have more complaints and take longer to resolve them. This illustrates how important factors influencing retention are issue resolution effectiveness and service responsiveness. Bad customer service experiences mount up quickly on subscription-based platforms and have a direct impact on cancellation decisions. Another significant factor contributing to churn is payment-related friction. Churn rates are significantly greater for customers who have billing problems, indicating that even little transactional disruptions can undermine client confidence. Furthermore, attrition differs within membership categories, suggesting that retention results may be influenced by perceived value, cost, or benefit differential.

Despite being less important than service characteristics, engagement indicators like watch duration and login frequency nevertheless play a significant role in churn prediction. Customers with lower levels of engagement may be more susceptible to dissatisfaction and switching behaviour, indicating that behavioural engagement and service experience interact. In terms of business recommendations, StreamFlex should give operational enhancements in customer service a priority in light of these findings. Strategic goals should include lowering the number of complaints and expediting the resolution process, maybe through proactive issue monitoring, automated ticket routing, and more staffing. Improving service effectiveness is probably going to help with retention right away. Secondly, in order to reduce payment friction, the business should optimize its billing system. Transactional barrier-related churn might be considerably decreased by using automatic payment failure notifications, flexible invoicing choices, and streamlined payment procedures.

Lastly, StreamFlex has to reevaluate its value proposition and membership structure. Price changes, feature differences, or bundled perks may increase perceived value if specific tiers show increased churn. By making the platform stickier, adding tailored content recommendations for people with low involvement can improve retention even further. All things considered, a strategy centred on value optimization, friction reduction, and service excellence is most likely to result in quantifiable and long-lasting increases in customer retention.